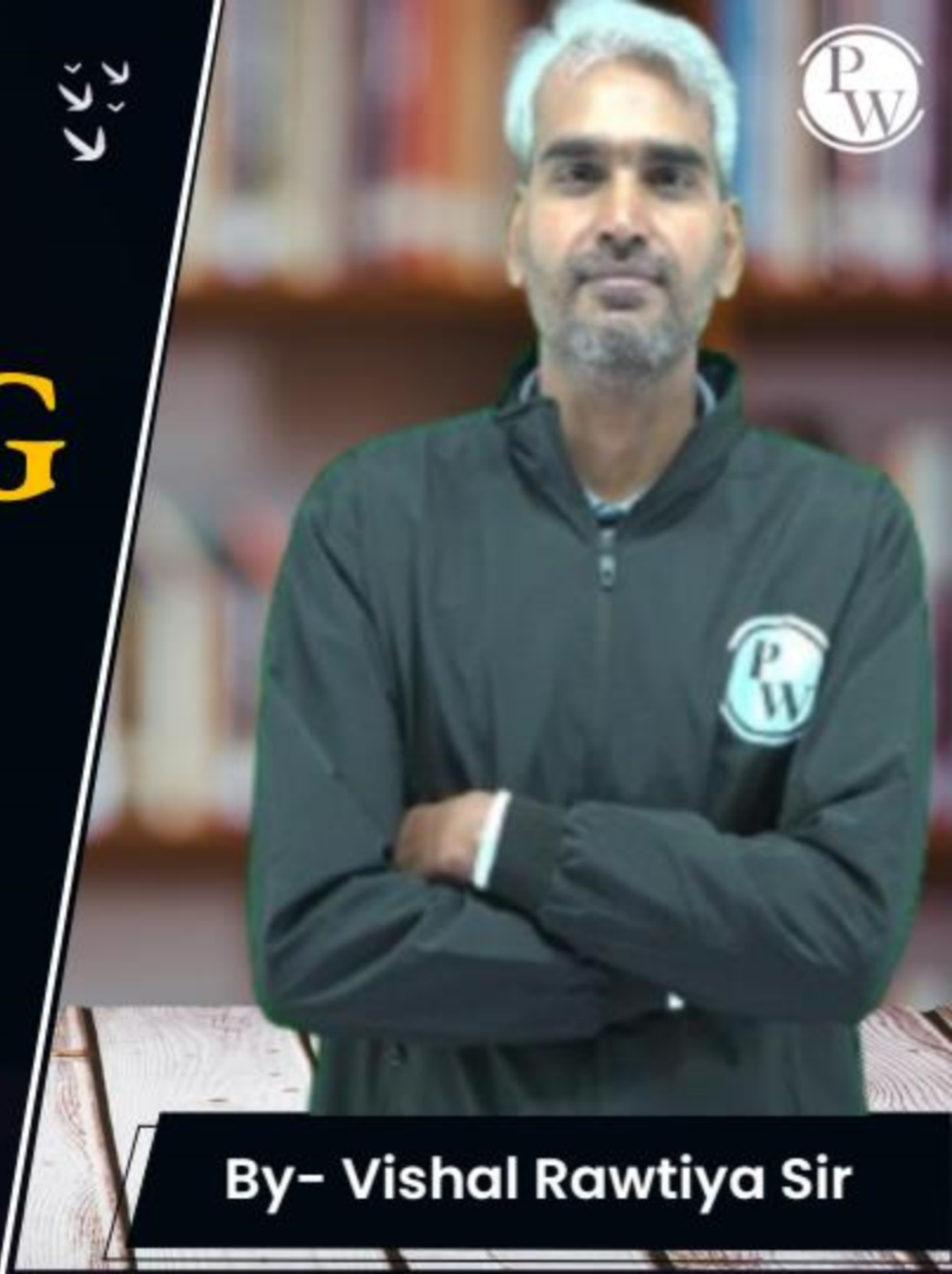
CS & IT ENGINEERING

Discrete Mathematics

Set Theory and Algebra

DPP-03 Discussion Notes



By- Vishal Rawtiya Sir

[NAT]

#Q. Let A= {a, b, c, d, e}. The cardinality of the equivalence relation on set A induced by partition {{a, c}, {d},{b, e}} of set A is _____

9

Ans

x x x

{a,c} {d} {b,e}

4 + 1 + 4 = 9

[NAT]

#Q. Let A is a finite set such that |A| = 5. Then cardinality of largest partial order relation possible on set A is _____ 15

Ans

if $|A| = n$, then

Size of largest partial order relation $= n + \frac{n^2 - n}{2}$

$= \frac{n^2 + n}{2}$

given $|A| = 5$

$\therefore = \frac{(5)^2 + 5}{2} = \frac{30}{2} = 15$

[NAT]

if $|A|=n$, then
no. of total order rel^n possible on set $A = n!$

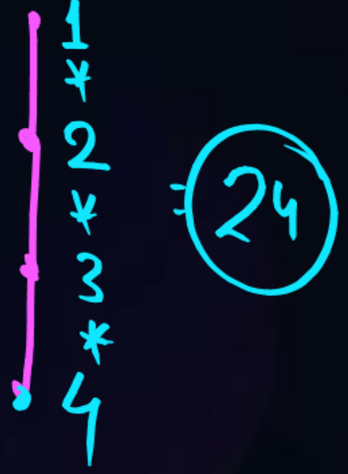
#Q. Let A = {a, b, c, d}

The number of total order relations possible on set A are 24

Ans

given $|A|=4$,

$\therefore$ No of total order Rel^n possible on set $A = 4! = 24$



$= 24$

[NAT]

#Q. Let A = {1, 2, 3, 4, 5, 6}.

The number of partitions of set A such that equivalence relations induced by those partitions will contain exactly 12 ordered pairs are ____35____

$|A| = 6$

$2+2+2 \Rightarrow \frac{{}^6C_2 * {}^4C_2 * {}^2C_2}{3!} = \frac{15 * 6 * 1}{6} = 15$

$1+1+1+3 : \frac{{}^6C_1 * {}^5C_1 * {}^4C_1 * {}^3C_3}{3!} = \frac{6 * 5 * 4 * 1}{6} = 20$

$15 + 20$

[MCQ]

R ⟵ Ref, Symm, trans.

S ⟵ Ref, Symm, Trans.

#Q. Let R and S are two equivalence relation on set A,

Which of the following is true?

A. R∪S is also an equivalence relation on set A. (Not necessary)

B. R∩S is also an equivalence relation on set A. (Always true)

C. Both R∪S as well as R∩S are equivalence relation on set A.

D. Neither R∪S nor R∩S are equivalence relation on set A.

If R & S are reflexive then both R∪S as well as R∩S are reflexive

If R & S are two symmetric Rel^n on set A, then both R∪S as well as R∩S are symmetric rel^n on set A.

$$R = \{(a,b), (b,a)\} \quad \cup \quad \{(a,a), (b,b), (c,c)\}$$

$$S = \{(a,a), (b,b), (c,c), (b,c), (c,b)\}$$

$$RUS = \{(a,a), (b,b), (c,c), (a,b), (b,a), (b,c), (c,b)\}$$

$(a,b) \in RUS$ & $(b,c) \in RUS$

but $(a,c) \notin RUS$

$\therefore$ not transitive

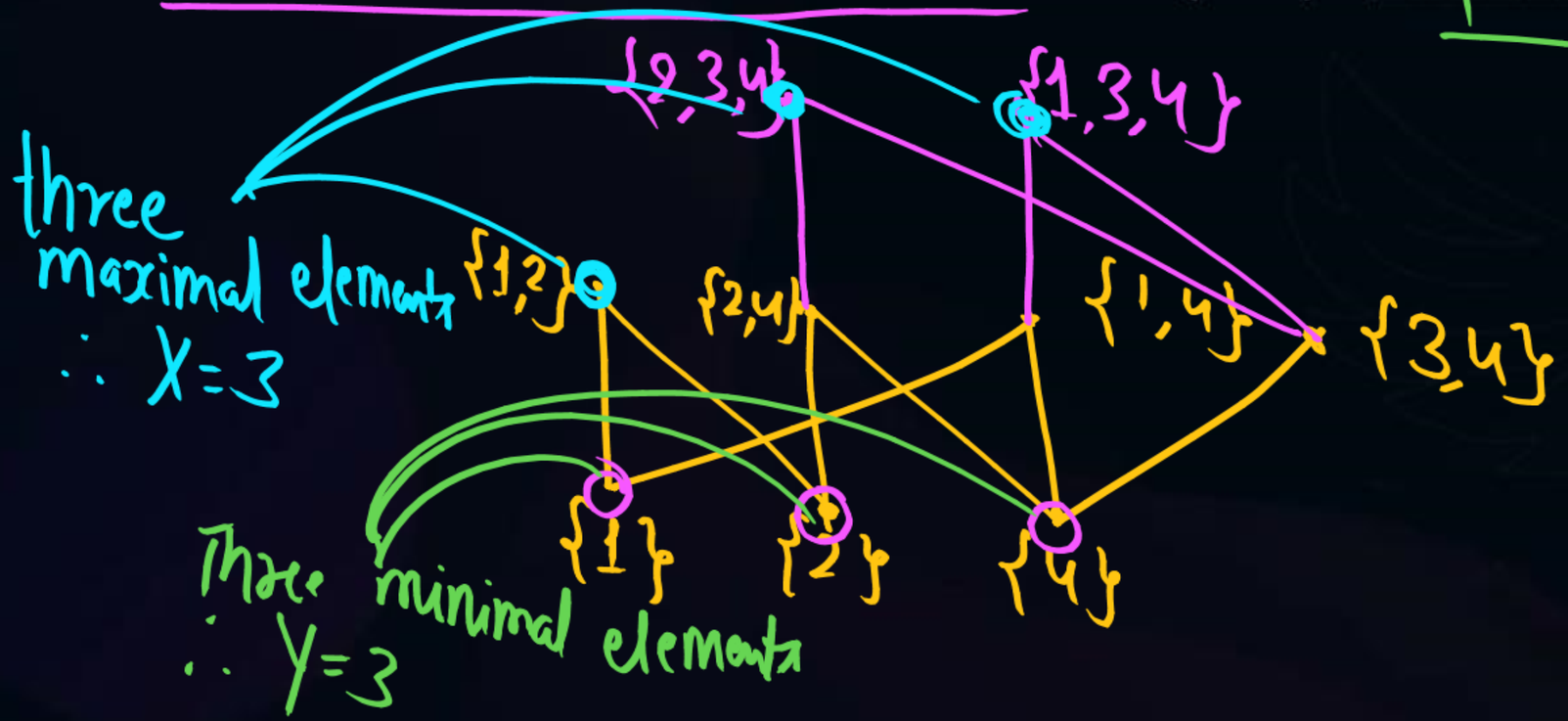
* If R & S are two transitive rel^n on set A, then $R \cap S$ is always transitive rel^n on set A, but $R \cup S$ need not be transitive rel^n on set A.

[NAT]

#Q. Consider the following POSET

$(\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}, \subseteq)$

Let X is the number of minimal elements in the POSET and Y is the number of maximal elements in the POSET. Then |X-Y| is $|3-3| = 0$ Ans

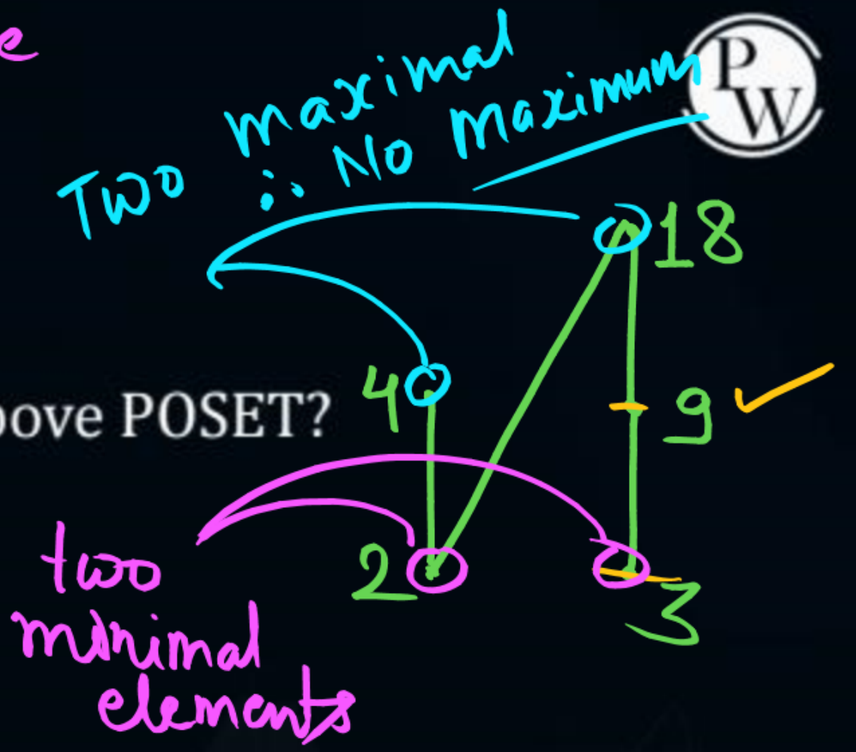


[MSQ]

Pair of elements which are not Comparable are = (2,3) (2,9) (3,4) (4,9) (4,18)

#Q. Consider the following POSET ({2, 3, 4, 9, 18}, |)

Which of the following is/are true with respect to above POSET?



A Maximum element does not exist in the POSET.

B Exactly four pairs of elements in the above POSET are not comparable.

five pair of elements are not Comparable

C Least upper bound of 3 and 9 is 18.

lub(3,9) = 9

D Number of minimal elements and number of maximal elements are same.

[MSQ]

#Q. Let A be a set with |A| = n, and let R be an equivalence relation on set A with |R| = r. Which of the following is TURE?

A. r – n will always be even.

B. r – n will always be odd.

C. r – n may be zero

D. r – n may be 1

All diagonal + Symmetric + Transitive

order Pairs

(a,b) (b,a) (b,c) (c,b) (a,c) (c,a)

are equal to the cardinality of set A = n

|R| = r = n + even value

|A| = n

r − n = (n + even value) − n = even value

w.r.t. diagonal Relⁿ

THANK - YOU